

CLASS 10 · CBSE · NCERT MATHEMATICS · NCERT SOLUTIONS

Chapter 10: Circles

Step-by-step solutions to the textbook exercises: 17 questions

Questions are numbered as in the textbook. Try each one yourself before reading the solution.

1 Exercise 10.1

Q1 State the number of tangents that a circle has.

SOLUTION

There is exactly one tangent at each point of a circle, and a circle has infinitely many points.

Answer: infinitely many

Q2 Complete each sentence: (i) A tangent meets a circle at exactly ___ point(s). (ii) A line that cuts a circle at two points is called a ___. (iii) At most ___ tangents to a circle can be parallel to one another. (iv) The point where a tangent touches the circle is known as the ___.

SOLUTION

(iii) Two parallel tangents touch the circle at the ends of a diameter, so there can be only two.

Answer: (i) one (ii) secant (iii) two (iv) the point of contact

Q3 A tangent PQ at a point P of a circle of radius 5 cm meets a line through the centre O at Q , with $OQ = 12$ cm. The length PQ is (A) 12 cm (B) 13 cm (C) 8.5 cm (D) $\sqrt{119}$ cm.

SOLUTION

The radius is perpendicular to the tangent at the point of contact, so $\angle OPQ = 90^\circ$.

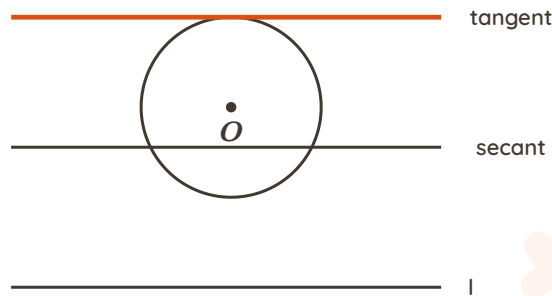
$$PQ = \sqrt{OQ^2 - OP^2} = \sqrt{144 - 25} = \sqrt{119} \text{ cm.}$$

Answer: (D) $\sqrt{119}$ cm

- Q4** Given a line, sketch a circle and then draw two lines parallel to the given line: one that is a tangent to the circle and one that is a secant.

SOLUTION

Draw the given line l and a circle with centre O . Draw a line parallel to l that cuts the circle at two points: this is the secant. Then draw another line parallel to l that just touches the circle at one point: this is the tangent.



The tangent and the secant are both parallel to the line l

The figure shows one such tangent and secant.

2 Exercise 10.2

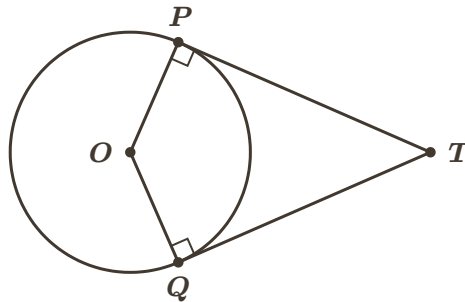
- Q1** A point Q is 25 cm from the centre of a circle, and the tangent from Q to the circle is 24 cm long. What is the radius? (A) 7 cm (B) 12 cm (C) 15 cm (D) 24.5 cm.

SOLUTION

The radius is perpendicular to the tangent at the point of contact, so radius = $\sqrt{25^2 - 24^2} = \sqrt{49} = 7$ cm.

Answer: (A) 7 cm

- Q2** TP and TQ are two tangents to a circle with centre O such that $\angle POQ = 110^\circ$. Then $\angle PTQ$ is (A) 60° (B) 70° (C) 80° (D) 90° .



SOLUTION

In quadrilateral $OPTQ$, $\angle OPT = \angle OQT = 90^\circ$ (radius $\perp$ tangent). The four angles add up to 360° .

$$\angle PTQ = 360^\circ - 90^\circ - 90^\circ - 110^\circ = 70^\circ.$$

Answer: (B) 70°

- Q3** Tangents PA and PB from a point P to a circle with centre O are inclined to each other at 80° . Then $\angle POA$ is (A) 50° (B) 60° (C) 70° (D) 80° .

SOLUTION

In quadrilateral $OAPB$, $\angle OAP = \angle OBP = 90^\circ$, so $\angle AOB = 180^\circ - 80^\circ = 100^\circ$.

$\triangle OAP \cong \triangle OBP$ (RHS: $OA = OB$, OP common), so OP bisects $\angle AOB$ and $\angle POA = 50^\circ$.

Answer: (A) 50°

- Q4** Show that the tangents at the two ends of any diameter of a circle are parallel to each other.

SOLUTION

Let AB be a diameter of a circle with centre O , and let PQ and RS be the tangents at A and B .

The radius is perpendicular to the tangent at the point of contact, so $OA \perp PQ$ and $OB \perp RS$.

That is, $\angle PAB = 90^\circ$ and $\angle ABS = 90^\circ$.

These are alternate interior angles made by the transversal AB with PQ and RS , and they are equal.

Hence $PQ \parallel RS$.

Q5 Show that the line drawn perpendicular to a tangent at its point of contact must pass through the centre of the circle.

SOLUTION

Let the tangent AB touch the circle with centre O at P . We know that $OP \perp AB$, because the radius through the point of contact is perpendicular to the tangent.

At a given point of a line, only one perpendicular to the line can be drawn. So the perpendicular to AB at P is the line OP itself.

Hence the perpendicular at the point of contact passes through the centre O .

Q6 The length of a tangent from a point A , 5 cm from the centre of a circle, is 4 cm. Find the radius of the circle.

SOLUTION

Let the tangent touch the circle at B , with centre O . Then $\angle OBA = 90^\circ$, $OA = 5$ cm and $AB = 4$ cm.

$$OB = \sqrt{5^2 - 4^2} = 3 \text{ cm.}$$

Answer: 3 cm

Q7 Two concentric circles have radii 5 cm and 3 cm. Find the length of the chord of the larger circle that touches the smaller circle.

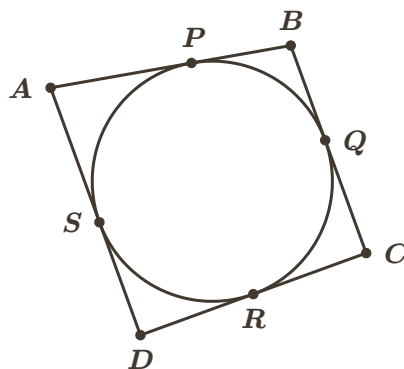
SOLUTION

Let the chord AB of the larger circle touch the smaller circle at M , with centre O . Then $OM \perp AB$, $OM = 3$ cm and $OA = 5$ cm.

$AM = \sqrt{5^2 - 3^2} = 4$ cm. The perpendicular from the centre bisects the chord, so $AB = 2 \times 4 = 8$ cm.

Answer: 8 cm

Q8 A quadrilateral $ABCD$ is drawn to circumscribe a circle. Prove that $AB + CD = AD + BC$.



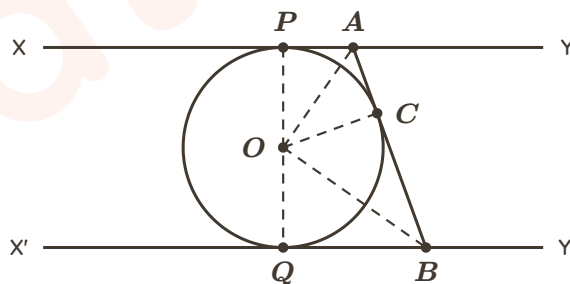
SOLUTION

Let the sides AB, BC, CD, DA touch the circle at P, Q, R, S . Tangents from an external point to a circle are equal, so $AP = AS, BP = BQ, CR = CQ$ and $DR = DS$.

Adding: $(AP + BP) + (CR + DR) = (AS + DS) + (BQ + CQ)$.

Hence $AB + CD = AD + BC$.

Q9 XY and $X'Y'$ are two parallel tangents to a circle with centre O . Another tangent AB , with point of contact C , meets XY at A and $X'Y'$ at B . Prove that $\angle AOB = 90^\circ$.



SOLUTION

Let XY touch the circle at P and $X'Y'$ touch it at Q . Join OC .

In $\triangle OPA$ and $\triangle OCA$: $OP = OC$ (radii), $AP = AC$ (tangents from A) and OA is common. So $\triangle OPA \cong \triangle OCA$ (SSS), and $\angle POA = \angle COA$.

In the same way $\triangle OQB \cong \triangle OCB$, so $\angle QOB = \angle COB$.

The tangents at P and Q are parallel, so PQ is a diameter and POQ is a straight line: $\angle POA + \angle COA + \angle COB + \angle QOB = 180^\circ$.

So $2(\angle COA + \angle COB) = 180^\circ$, that is $\angle COA + \angle COB = 90^\circ$.

Hence $\angle AOB = 90^\circ$.

Q10 Two tangents are drawn to a circle from a point outside it. Prove that the angle between the tangents and the angle that the chord of contact makes at the centre add up to 180° .

SOLUTION

Let PA and PB be the tangents from P to a circle with centre O . Then $\angle OAP = \angle OBP = 90^\circ$ (radius $\perp$ tangent).

In quadrilateral $OAPB$: $\angle APB + \angle AOB + 90^\circ + 90^\circ = 360^\circ$.

Hence $\angle APB + \angle AOB = 180^\circ$, so the two angles are supplementary.

Q11 Prove that a parallelogram circumscribing a circle is a rhombus.

SOLUTION

Let $ABCD$ be a parallelogram circumscribing a circle. Then $AB = CD$ and $AD = BC$ (opposite sides of a parallelogram).

For any quadrilateral circumscribing a circle, $AB + CD = AD + BC$ (proved in Q8).

So $2AB = 2AD$, that is $AB = AD$. Then $AB = BC = CD = DA$.

A parallelogram with all sides equal is a rhombus.

Q12 A triangle ABC circumscribes a circle of radius 4 cm. The point of contact D divides BC into $BD = 8$ cm and $DC = 6$ cm. Find AB and AC .

SOLUTION

Let the circle touch AB at F and AC at E . Tangents from a point are equal: $BF = BD = 8$ cm, $CE = CD = 6$ cm, and let $AF = AE = x$ cm.

Then $AB = 8 + x$, $AC = 6 + x$, $BC = 14$, and the semi-perimeter is $s = 14 + x$.

By Heron's formula, area = $\sqrt{s(s-a)(s-b)(s-c)} = \sqrt{(14+x) \times x \times 8 \times 6} = \sqrt{48x(14+x)}$.

Joining the centre O to the vertices splits the triangle into three triangles of height 4 cm: area = $\frac{1}{2} \times 4 \times (AB + BC + CA) = 4(14 + x)$.

So $48x(14 + x) = 16(14 + x)^2$, which gives $3x = 14 + x$ and $x = 7$.

Answer: $AB = 15$ cm and $AC = 13$ cm

Q13 A quadrilateral is drawn around a circle so that all four of its sides touch the circle. Prove that any two opposite sides make angles at the centre that add up to 180° .

SOLUTION

Let $ABCD$ touch the circle with centre O at P, Q, R, S on AB, BC, CD, DA (as in the figure of Q8). Join O to the four vertices and the four points of contact.

$\triangle AOP \cong \triangle AOS$ (SSS: $OP = OS$, $AP = AS$, OA common), so $\angle AOP = \angle AOS$. In the same way $\angle BOP = \angle BOQ$, $\angle COQ = \angle COR$ and $\angle DOR = \angle DOS$.

The eight angles at O add up to 360° , so $2(\angle AOP + \angle BOP + \angle COR + \angle DOR) = 360^\circ$.

That is $(\angle AOP + \angle BOP) + (\angle COR + \angle DOR) = 180^\circ$, so $\angle AOB + \angle COD = 180^\circ$. Similarly $\angle BOC + \angle AOD = 180^\circ$.

Hence opposite sides subtend supplementary angles at the centre.

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